

WEEK 2 DELIVERABLE REPORT: STAKEHOLDER MAP + BUSINESS CASE DRAFT + AUDIT LOG

Precision Agricultural Drone Spraying Coordination Platform for the Mekong Delta (AgriUAV Platform)

Project Metadata & Overview

Course	ITA301 — Information System Analysis & Design
Academic Term	Summer 2026
Deliverable Item	W2 — Stakeholder Map + Business Case Draft + Audit Log (Submission Deadline: 2026-05-31)
System Domain	Agricultural UAV — Precision Drone Spraying Coordination in the Mekong Delta
Course Instructor	Mr. Tran Thanh Nguyen
Project Team & Roles	- Dang Thanh Cong — Project Manager - Nguyen Thi Thanh Thao — Business Analyst - Le Van Minh — System Analyst - Tran Anh Khoa — System Designer - Dang Vo Thanh Hieu — QA / Presenter
Document Version	v1.1 — Official W2 Document

1. Comprehensive Stakeholder Map

Based on the ecosystem decomposition of the digital agriculture infrastructure within the Mekong Delta, the project team identified and categorized key stakeholders into three groups: Primary, Secondary, and Regulatory. Two critical stakeholders omitted by the initial AI output were supplemented by the team based on field-level regulatory research, and are marked with (*).

Classification	Stakeholder Group	Core Role & System Expectations	Power (1–5)	Interest (1–5)	Strategic System Action
Primary	Cooperative Management (HTX)	Optimize fleet scheduling and dispatching, reduce operational costs by 15–20%, and obtain verified digital records of pilot activity and pesticide usage.	5	5	Web Dashboard for real-time fleet scheduling, shift allocation, and compliance reporting.
Primary	Operators / Drone Pilots	Receive clear flight assignments, view digital field boundary maps, navigate safely away from power lines, and log verified flight hours.	2	5	Lightweight Mobile App with GPS field navigation, check-in/out confirmation, and automated flight log generation.
Primary	Rice Farmers	Book spray services with minimum effort (≤5 min), track drone spray status in real-time, and verify actual pesticide volume dispensed.	3	4	Automated SMS/Zalo push notifications providing real-time spray order status updates.

Classification	Stakeholder Group	Core Role & System Expectations	Power (1–5)	Interest (1–5)	Strategic System Action
		Avg. age ~51 (Nguyen 2026), limited IT experience.			
Primary	Agricultural Engineers / Agronomists	Define and validate spray formulas per crop type and pest category, manage approved pesticide registry, and confirm correct dosage application.	3	4	Formula management module with pesticide registry integration aligned to Plant Protection Dept approved chemical list.
Secondary	Agricultural Exporters (Trung An, Minh Phu)	Trace full pesticide application history to demonstrate low residue compliance for EU/US export standards (VietGAP/GlobalGAP).	3	4	Exportable spray history report module formatted to VietGAP/GlobalGAP traceability standards.
Secondary	Drone Manufacturers (DJI, XAG)	Expand market reach through platform compatibility; benefit from telemetry integration via MAVLink SDK.	2	3	MAVLink telemetry API integration roadmap (planned for post-MVP release).
Secondary	Banks / Agricultural Credit Institutions	Use verified seasonal production data as a basis for agricultural lending risk assessment.	2	2	Seasonal operation summary export with verified flight logs usable as credit evidence.
Secondary (*)	Pesticide Retailers	Supply approved chemical inputs to HTX cooperatives; inventory sync prevents unapproved pesticide use. [Supplemented by team — AI omitted this actor]	2	2	Chemical inventory sync API connected to approved chemical registry for real-time stock validation.
Regulatory	Plant Protection Dept. (Cục Trồng trọt & BVTV)	Approve pesticide categories permitted for UAV application; monitor seasonal compliance reporting and chemical residue traceability.	5	3	Automated compliance export (PDF/Excel) containing drone IDs, pilot certificates, and pesticide usage logs per season.
Regulatory (*)	Operations Dept. / General Staff (Cục Tác chiến — Bộ Tổng Tham mưu)	Sole military authority responsible for UAV flight permits and drone identification registration under Decree 288/2025/ND-CP. [Supplemented by team — AI omitted this actor]	4	2	Store and display device tracking UUIDs and certified pilot license numbers in compliance reports submitted per Decree 288.
Regulatory	Provincial Authorities (UBND / Sở NN&PTNT ĐBSCL)	Grant operational licenses for drone service providers at the provincial level; receive periodic area-sprayed reports.	3	3	Periodic provincial spraying area reports exportable for submission to Ministry of Agriculture.

(*) Supplemented by team — not identified in initial AI stakeholder output. Verified against Decree 288/2025/ND-CP (Operations Dept.) and domain knowledge of cooperative supply chains (Pesticide Retailers).

2. Current Process (AS-IS) & Measurable Pain Points

The existing operational flow within agricultural cooperatives follows five sequential phases. Field evidence and documented incidents expose critical pain points at each stage that require digital infrastructure solutions.

Seq	Operational Phase	As-Is Manual Workflow	Measurable Localized Pain Points
1	Demand Intake	Farmers call or text the HTX manager via Zalo to request drone spraying, providing a rough verbal estimate of land area.	Lack of GPS Coordinates: Field plots are frequently misidentified due to absent digital boundary data, leading to accidental spraying on adjacent parcels and neighbor disputes.
2	Scheduling & Dispatch	The manager manually records dates in paper logbooks or a shared Excel file, then calls pilots individually to check availability.	Double-Booking & Deadhead Flights: Manual scheduling causes persistent booking conflicts and misallocated dispatch routes, inflating operational costs by 15–20% per season.
3	Flight Execution	Pilots operate drones based on visual memory or standalone hardware flight apps with no centralized telemetry.	Unverified Operator Activity: Absence of centralized telemetry verification allows operators to under-dose chemical mixtures or fabricate flight records to claim fuel and pesticide allowances without detection.
4	Monitoring & Control	No real-time operational dashboard exists. Management remains entirely blind until the pilot verbally reports completion by phone.	CRITICAL SAFETY RISK: Absence of geofencing has resulted in drone collisions with 110kV power transmission lines (Long An Province — 76,000 customers affected, Dan Viet 2024). Off-target chemical drift causes exposure to neighboring organic crops.
5	Reconciliation & Reporting	At month-end, HTX managers aggregate paper invoices to bill farmers and compile handwritten regulatory reports for submission.	Compliance Failure: Error-prone manual processes cannot satisfy Decree 288/2025/ND-CP, which mandates digital logs of unique drone IDs and pilot credentials. No verifiable audit trail exists.

3. Business Case Draft & Economic Evaluation

3.1. Ecosystem Gap Analysis — 3-Layer Model

The Agricultural UAV ecosystem in Vietnam is structured across three distinct layers. The analysis below demonstrates the specific market gap that AgriUAV Platform is designed to occupy:

Technology Layer	Current Ecosystem Market State	AgriUAV Platform Intervention
Hardware Layer	Dominated by mature global manufacturers (DJI Agras ~80%, XAG ~15%, domestic ~5%). Market is highly commoditized and saturated. Hardware availability is not a bottleneck.	Integration via mobile GPS telemetry at MVP stage. Planned native integration via MAVLink protocols in future versions.
Service Layer	Local cooperatives (HTX) provide commercial spraying services (~150,000 VND/ha) but operate on entirely manual, fragmented processes (Zalo/Excel/paper logbooks).	Provides the end-to-end digital coordination fabric to automate internal scheduling, dispatch, and compliance workflows for these service providers.
Platform Layer	COMPLETELY VACANT — No dedicated, localized IS for fleet coordination, pilot	AgriUAV Platform fills this strategic gap, serving as the enterprise-grade

Technology Layer	Current Ecosystem Market State	AgriUAV Platform Intervention
	dispatch, scheduling, or regulatory compliance logging currently exists in Vietnam.	operational coordination system for drone service cooperatives.

3.2. Market Sizing Estimates (TAM / SAM / SOM)

Market sizing is derived from official field statistics and empirical research to ensure figures are verifiable rather than estimated:

Segment	Drone Count & Data Source	Target Customer Profile	Strategic Rationale
TAM (Total Addressable Market)	~4,000 agricultural drones active across 1.5 million ha of rice cultivation land in the Mekong Delta.	Full Mekong Delta drone fleet	No digital platform in market = full addressable market
SAM (Serviceable Addressable Market)	1,215 active drones in 3 priority provinces: Kien Giang (690), An Giang (265), Dong Thap (260). Source: Thanh Nien, June 2025.	HTX and commercial drone service operators	Operationally reachable in Year 1 rollout
SOM (Serviceable Obtainable Market)	200 drones managed by pilot-partner cooperatives linked with Can Tho University research network.	Strategic pilot HTX partners	Direct partnership channel for initial deployment

Monetization Model: B2B SaaS licensing at 500,000 VND / drone / month. Projected annual recurring revenue from SOM segment: $200 \times 500,000 \times 12 = 1,200,000,000$ VND/year.

3.3. Cost-Benefit Analysis (Reference: 1 HTX — 10 drones, 500 ha/season)

The following cost-benefit breakdown uses a representative mid-size cooperative as the reference unit. All figures are conservative estimates derived from operational benchmarks reported in Nguyen et al. (2026) and field reporting:

Cost Category	AS-IS (Manual Operations)	TO-BE (With AgriUAV Platform)	Estimated Savings
Pesticide wastage (imprecise spraying)	~45 million VND/season	~31.5 million VND/season	–30% = 13.5M VND
Drone fuel (duplicate flights, wrong zones)	~12 million VND/season	~9.6 million VND/season	–20% = 2.4M VND
Coordination & reporting labor	~8 million VND/season	~2 million VND/season	–75% = 6.0M VND
Legal penalty risk (Decree 288 non-compliance)	Unquantified — ongoing exposure	Significantly reduced	Compliance built-in

Cost Category	AS-IS (Manual Operations)	TO-BE (With AgriUAV Platform)	Estimated Savings
TOTAL ESTIMATED SAVINGS / SEASON			~21.9 million VND

Platform operating cost estimate: ~1,200,000 VND/month (cloud infrastructure + API services). For a 3-month rice season, platform cost = 3,600,000 VND vs. estimated savings of 21,900,000 VND → Positive ROI from the first season of deployment.

3.4. Preliminary Cost-Benefit Metric Overview

Key Metric	Pre-System Baseline (AS-IS)	Post-System Intervention Target
Operational Fleet Cost / Season	Baseline manual inefficiencies; costs inflated 15–20% through scheduling waste and deadhead flights.	15–20% cost reduction via elimination of duplicate bookings and route optimization.
Compliance Report Turnaround	Requires 3–5 business days of manual data entry and paper compilation per reporting batch.	Under 1 hour via automated compliance export engine (PDF/Excel).
Critical Flight Safety Events	Zero visibility; no geofencing — highly prone to infrastructure collision incidents.	Proactive geofencing blocks drone entry into registered high-voltage transmission line corridors.
Operator Activity Verification	Entirely unverifiable; exposed to under-dosing and inflated flight hour claims.	GPS-based field check-in/out protocols enforce verified physical presence at spray sites.
Potential Annual Revenue (SOM)	Not applicable — no platform revenue model exists.	200 drones × 500,000 VND × 12 months = 1,200,000,000 VND/year (B2B SaaS model).

3.5. Business Value Summary

- Operational Efficiency: 15–20% reduction in per-season operational costs — validated by Nguyen et al. (2026), N=940 farms surveyed.
- Regulatory Compliance: Full alignment with Decree 288/2025/ND-CP — eliminates legal liability from non-compliant manual logging.
- Flight Safety: Automated geofencing and power-line hazard alerts — directly address documented safety incidents (Long An 110kV collision, Dan Viet 2024).
- Environmental Impact: Targeted dosage delivery reduces agrochemical wastage by 20–30%; water usage reduction up to 70% versus conventional spraying (Vietnam News, 2026).
- First-Mover Advantage: Platform layer is entirely unoccupied in the Vietnamese agricultural UAV market — no domestic competitor currently operates at this coordination layer.

4. AI Artifact Audit Log — v1.1

This audit log documents the iterative cycle of prompt engineering, human critique, and systematic refinement executed by Team 3 during Week 2. Entries continue from #04 in sequence with the Week 1 log (entries #01–#03). Each entry highlights the Human Delta — the critical scholarly adjustments made to correct generic AI gaps and anchor the analysis in local Vietnamese field realities.

No .	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta	Verification
04	Decomposition (Stakeholder Mapping)	Claude	"Project: AgriUAV Platform for drone spraying in Mekong Delta. List all stakeholders, classify as Primary/Secondary/Regulatory, and outline core roles and expectations for each."	Suggested 6 general groups: Farmers, HTX, Pilots, Drone Manufacturers (DJI), Software Developers, and Local Government — totaling ~12 stakeholders.	☒ Accepted 5 core stakeholder groups. ☒ Removed 'Software Developer Team' — this is the project team, not an external stakeholder. ☒ SUPPLEMENTED 2 critical actors AI omitted: • Operations Dept. / General Staff — sole authority for UAV flight permits under Decree 288/2025 (confirmed via Vietnam Naval News). • Pesticide Retailers — essential for inventory/compliance integration in cooperative supply chains. ☒ Split generic 'Government' into 2 specific regulatory bodies: Plant Protection Dept. + Operations Dept.	Operations Dept. authority confirmed: Vietnam Naval News — 'all flight permit records must be submitted to Cuc Tac chien, causing administrative delays.'
05	Decomposition (AS-IS Process Analysis)	Claude	"Decompose the AS-IS business process of an agricultural cooperative in Vietnam executing drone spraying services. List each step, the actor responsible, and pain points at each stage. Context: currently managed via Zalo/Excel."	Proposed a basic 4-step generic workflow: Intake → Dispatch → Execution → Reporting. Identified high-level pain points: lack of tracking, slow report compilation.	☒ Accepted 4-step structure as baseline. ☒ Expanded to 5 steps — added 'Monitoring & Control' as a distinct phase (AI merged this into Execution). ☒ Localized pain points with field-verified data: • Phase 3: Operators may under-dose chemicals or inflate flight hours; no audit trail — AI completely missed this risk. • Phase 4: Integrated the documented 110kV power line collision in Long An (Dan Viet 2024) as concrete evidence for geofencing necessity.	110kV incident: Dan Viet (2024) — drone collision in Long An Province, 76,000 customer blackout. Decree 288/2025/ND-CP text: Article on drone ID and pilot credential logging requirements.

No .	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta	Verification
					Phase 5: Linked compliance failure directly to Decree 288/2025/ND-CP specific requirements.	
06	Process / Algorithm (Business Case Structure)	Claude	"Provide an industry-standard business case structure for an IS research project, including a preliminary ROI template and a formal problem statement formulation suitable for a university-level IS analysis project."	Provided a standard 5-part structure (Context → Problem → Impact → Solution → Scope) and a generic theoretical ROI formula: (cost saved / investment) × 100%.	☒ Adopted structural blueprint as foundation. ☒ Applied Mekong Delta data localization: - Impact metric: 15–20% operational cost increase (Nguyen et al. 2026, N=940) — not AI's generic estimate. - Market size: 1,215 drones across 3 provinces from Thanh Nien June 2025 — hard figures replacing AI's generic framing. - Added HTX-level Cost-Benefit table (not in AI output): pesticide –30%, fuel –20%, labor –75% = ~21.9M VND savings/season vs. 3.6M VND platform cost → positive ROI from Season 1. - Established 45% unmanaged area ratio as the addressable market segment (Nguyen 2026).	15–20% operational cost: Nguyen et al. (2026). Drone count by province: Thanh Nien (June 2025). Cost-benefit figures: team derivation from published operational benchmarks, conservative estimates.

5. Verified Academic & Regulatory References

[1] Nguyen, D.L. et al. (2026). "The Effect of Farm Size on the Decision to Adopt Digital Technology: The Case of Unmanned Aerial Vehicles in Rice Production in Vietnam." Research on World Agricultural Economy. (N=940). <https://journals.nasspublishing.com/index.php/rwae/article/view/2358>

[2] Government of Vietnam. Decree No. 288/2025/ND-CP — Regulatory Framework for Unmanned Aerial Vehicle Flights and Light-UAV Digital Registry (Effective: November 05, 2025). <https://thuvienphapluat.vn/van-ban/Giao-thong-Van-tai/Nghi-dinh-288-2025-ND-CP-quan-ly-tau-bay-khong-nguoi-lai-va-phuong-tien-bay-khac-679996.aspx>

[3] Thanh Nien Newspaper. (June 2025). Province-Level Statistical Breakdown of Active Agricultural Drone Fleets in the Mekong Delta Region. <https://thanhvien.vn/bung-no-uav-nong-nghiep-o-mien-tay-185250617232615926.htm>

[4] Dan Viet Digital News. (2024). Investigation Report: Agricultural Drone Incident Involving 110kV Transmission Line Contact, Causing Outages for 76,000 Customers in Long An Province. <https://tuoitre.vn/drone-phun-thuoc-tru-sau-va-vaoday-110kv-gay-mat-dien-o-5-huyen-tai-long-an-20241014173928997.htm>

- [5] Vietnam Naval News. (2024). Management of Unmanned Aerial Systems and Flight Permit Processes via the Operations Department (Cuc Tac chien). <https://baohaiquanvietnam.vn/tin-tuc/quan-ly-thiet-bi-bay-khong-nguoi-lai-trong-nong-nghiep-khong-de-cong-nghe-tro-thanh-hiem-hoa>
- [6] Vietnam News. (2026). Use of Agricultural Drones on the Rise in the Mekong Delta. <https://vietnamnews.vn/society/1688198/use-of-agricultural-drones-on-the-rise-in-the-mekong-delta.html>
- [7] Lao Dong Newspaper. (2025). Digital Transformation Gaps in Mekong Delta Cooperatives: The Unresolved Software Problem. <https://laodong.vn/xa-hoi/drone-nong-nghiep-va-bai-toan-so-hoa-o-dbscl-1498970.lido>
- [8] CropLife Vietnam. (2025). Forum: Making Drones an Effective Assistant for Farmers. <https://croplifevietnam.org/toa-dam-dua-drone-tro-thanh-tro-thu-dac-luc-cua-nha-nong.html>